Beginner's Guide to the OULAD Project Notebooks

i Oct 24, 2026 · @what are you

Nine notebooks, run in order, turn raw student records into an early-warning model; this guide explains what each one does in plain words.

The big picture

The project finds students who are in trouble after the first four weeks of a course, so that a tutor can help them early. It uses real but anonymous records from the Open University in the UK.

Think of the nine notebooks as a factory line. Each one does one job, then hands its result to the next. A notebook is a file that mixes explanations, code and results; you read it from top to bottom.

Notebook	The question it answers	What you get
01 Framing	What are we asking, and where does the data come from?	A clear question and a map of the data
02 Master table	How do we turn seven messy files into one clean table?	One table with 27,553 students
03 Exploring	What is in the data, and what is wrong with it?	Charts, a list of problems, first findings
04 Features	Which facts about a student help to predict trouble?	18 chosen facts (features)
05 Clustering	Do students fall into types?	Three study styles, plus "no activity"
06 Rules and tests	Which combinations of facts go with trouble?	173 rules, tests and maps
07a Model	Can we predict trouble in a new year, and is it fair?	The prediction model and its exam result

Notebook	The question it answers	What you get
07b Pitfalls	What can silently go wrong?	Warnings and a monitoring plan
07c Time and A/B	Can we forecast platform use, and test whether help works?	Forecasts and a trial design

Run them in this order from the `notebooks` folder. Each section below says what the notebook does, why it matters, and what to look for in its results.

Notebook 01: Framing and provenance

In one sentence: it decides what question we are asking and checks that the data is what we think it is.

What it does

- Writes down the question (who is at risk after four weeks?) and four smaller questions.
- Says where the data comes from: the Open University, downloaded from the UCI repository, with its licence.
- Loads the six small tables and counts their rows.
- Checks that every ID that should be unique really is unique (for example, one row per student per course).
- Finds a typo in the data's own documentation: it says 32,953 students in one place and 32,593 in another. The data has 32,593.

Why it matters. Before any analysis you must know what one row means. Here, one row is one student in one course run. The same person can appear several times: 28,785 different people fill 32,593 rows.

What you get: a clear question, a map of the seven tables, and the facts about the data that later notebooks rely on.

Look for: the table that lists each file, its number of rows and what one row stands for.

Notebook 02: The master table

In one sentence: it combines the seven files into one clean table with one row per student, using only what was known by day 28.

What it does

- Reads the files, treating the symbol `?` as "missing".
- Chooses the prediction day: day 28, about the end of week four. It draws a chart of when students withdraw.
- Removes the 5,040 students who left before day 28. Their result is already known, so keeping them would be cheating.
- Counts what each student did in the first 28 days: clicks per week, days active, which kinds of material they used.
- Adds early assessment facts: what they handed in and their scores.
- Cleans labels and sets the target: **at risk = ended with Fail or Withdrawn**.
- Writes every change into a log, so nothing is hidden.

Why it matters. The rule of the whole project is: never use information from the future. This notebook builds that rule into the data.

What you get: `oulad_master_day28.csv` (27,553 students), a data dictionary and a transformation log.

Look for: the chart of withdrawal timing, and the check that students with more clicks are less often at risk (62.5% down to 27.1%).

Notebook 03: Exploring and checking the data

In one sentence: it looks at the data from every side and makes a list of what is wrong with it.

What it does

- Runs a quality audit: missing values, impossible values, rare categories, extreme values and anything that could leak the answer.
- Tests two worries: do the duplicate click rows matter, and what about 93 withdrawn students with no exit date? (Neither changes the results.)
- Summarises each variable on its own, with histograms and tables.
- Compares pairs of variables, such as clicks against active days, or deprivation against risk, and measures how strong each link is.
- Checks whether a pattern still holds inside each module, so it is not just a mix-up of courses.

- Finds a coding problem: one assessment is scored only 100 or 1, so it is a tick box, not a real mark.

Why it matters. You cannot trust a model built on data you have not examined. Notice that the notebook reports how *big* each effect is, not only whether it exists; with this many students, almost everything is "significant".

What you get: an audit table, effect-size tables and nine charts.

Look for: the chart of at-risk rate by deprivation band (56% in the most deprived band, 35% in the least).

Notebook 04: Building and choosing features

In one sentence: it builds the facts (called features) that the model will use, then keeps the best 18 out of 48.

What it does

- Fixes the score problem: each score becomes a rank among students who did the same assessment (top 10%, middle, and so on).
- Makes engagement *relative*: "is this student more active than other students in the same course?" This matters because some courses are ten times busier than others.
- Adds shape facts, such as whether activity is rising or falling and the longest break.
- Splits the data into a **training set** (2013) and a **test set** (2014). Everything below uses only the training set.
- Removes features that say the same thing twice, ranks the rest in three different ways, and picks how many to keep by experiment.
- Compares groups of features to see how much accuracy each group adds.

Why it matters. Good facts usually help more than a clever model. Choosing features on the training years only avoids cheating on the test.

What you get: `oulad_features_day28.csv` and the list of 18 chosen features.

Look for: the chart of accuracy by feature group: module alone is barely better than guessing, assessment behaviour adds the most.

Notebook 05: Clustering students into groups

In one sentence: it sorts students into groups by how they study, without ever telling the computer who failed.

What it does

- Picks 12 facts about study behaviour and leaves out the outcome, the student's background and their marks.
- Tries four grouping methods: k-means, hierarchical clustering, DBSCAN and a Gaussian mixture.
- Decides how many groups to use with rules written down in advance, and checks that the groups stay the same when the data is resampled.
- Describes each group and measures its risk afterwards.
- Applies the groups to the 2014 students (without looking at their results) to see if they still make sense.

A mistake the notebook catches. The first attempt produced groups that were really just the courses. The fix was to compare each student only with students in the same course. The notebook keeps both versions so you can see the difference.

Why it matters. Three simple groups are easy for tutors to use: low, in-and-out, and steady. The honest finding is that students form a smooth range, not neat types, so the groups are useful labels, not natural categories.

What you get: the group of each student and a table of group profiles.

Look for: the bar chart of risk by group: no activity 81%, low 63%, in-and-out 47%, steady 29%.

Notebook 06: Rules and category tests

In one sentence: it looks for combinations of facts that go together with trouble, and tests whether those links are real.

What it does

- Turns facts into simple categories, such as deprived, middle or affluent, or "missed an early assessment".
- Runs tests that measure how strongly each category is linked to being at risk (Cramér's V runs from 0, no link, to 1, perfect link).

- Checks that a link is not caused by the course mix, by comparing students inside the same course.
- Searches for rules of the form "if A and B, then at risk", and keeps only the 173 that are strong, large enough and not just a copy of a simpler rule (from 5,123 candidates).
- Draws maps (correspondence analysis) that show which categories sit close to which results.

Why it matters. Rules are easy to explain to non-experts: for example, a student who missed an assessment in the first four weeks ended up at risk 87.6% of the time. But a computer that tries thousands of rules will find false patterns, which is why the notebook filters so hard.

What you get: the rule list, test results and a ready-made "who to contact first" list.

Look for: the chart of the strongest rules, and the finding that gender has no link with risk.

Notebook 07a: The prediction model and its exam

In one sentence: it trains a model on 2013 and tests it once, like a final exam, on 2014.

What it does

- Writes the rules of the test first (which features, how to pick the model, how many students to flag) so nothing can be changed after seeing the results.
- Tries different settings by cross-validation, which means practising on 2013 only. It tests two kinds of model: logistic regression (simple) and gradient boosting (more flexible).
- Gives the final exam on 2014 and reports how good it is (AUC), whether its percentages are honest (calibration), and how it does in each course.
- Checks fairness: are at-risk students found equally often in every group (by deprivation, education, gender, disability, age)? It also tests a version of the model that does not see education or deprivation.
- Re-tests the rules and groups from earlier notebooks on 2014, and asks how early in the course the model can start working.

Why it matters. A model is only useful if it works on students it has never seen. This is also where the project decides which model to recommend.

What you get: the 2014 predictions and a record of the final model settings.

Look for: AUC 0.78 (1.0 is perfect, 0.5 is guessing), and the fairness chart showing different recall by group.

Notebook 07b: Imbalance, leakage and drift

In one sentence: it tests three things that can quietly ruin a model that looks good on paper.

What it does

- **Class imbalance:** some outcomes are rare (only about 17% withdraw). It compares the usual fixes (weights, copying or deleting rows, SMOTE, moving the cut-off) and finds that moving the cut-off works as well as any of them.
- **Data leakage:** it adds forbidden information on purpose, such as clicks from after day 28, and shows how much the score is inflated (from 0.78 to 0.86). It also shows several other ways to cheat by accident, such as choosing features before testing.
- **Data drift:** it compares 2013 and 2014 to see what changed (which courses, deadlines, codes) and whether the link between features and outcome changed. It did not.
- Ends with a monitoring plan: what to check each week, and when to retrain.

Why it matters. Leakage and drift are the usual reasons a model that worked in the lab fails in real life. The notebook measures them instead of only describing them.

What you get: a leakage table, a drift table and a monitoring plan.

Look for: the table that shows how much each kind of cheating inflates the score, and the chart of which inputs moved between years.

Notebook 07c: Time series and A/B test design

In one sentence: it forecasts how busy the platform will be, and plans a fair test of whether contacting students actually helps.

Part A, time series (data ordered by day)

- Adds up the clicks per day for each of the 22 course runs.
- Splits each curve into a slow trend, a weekly cycle and spikes, and finds that deadlines cause the biggest spikes (activity is up to three times higher the day before a due date).
- Forecasts days 28 to 55 from the first 28 days with several methods and compares their errors. Using last year's course as a template works best.

Part B, A/B test design

- Works out how many students are needed to see a real effect (about 2,360 per group to detect a 3.7 point drop in risk).
- Simulates thousands of imaginary trials to check that the test is reliable.
- Shows why simply comparing students who were contacted with those who were not gives the wrong answer: contacted students were already the riskiest.
- Ends with a written experiment plan: groups, measures, safeguards and ethics.

Why it matters. Predicting who is at risk does not prove that helping them works. Only a fair experiment can show that.

What you get: forecast results, a power chart and the experiment plan.

Look for: the forecast chart, and the table of how big the trial must be.

Word list

Word	Plain meaning
Feature	One fact about a student that the model can use, such as total clicks in the first 28 days
Target	The thing we want to predict: here, "at risk" (Fail or Withdrawn)
Course run (module-presentation)	One course in one start month, for example module FFF starting in October 2014
Cutoff (day 28)	The day we make the prediction; we may only use what happened before it
Training set and test set	Training: the years the model learns from (2013). Test: a later year it has never seen (2014)
Cross-validation	Practising on the training data by repeatedly hiding a part of it and predicting that part
AUC	A score from 0.5 (guessing) to 1.0 (perfect) for how well the model puts at-risk students above the others

Word	Plain meaning
Calibration	Whether a predicted 40% risk really means about 40 in 100 students end up at risk
Precision and recall	Precision: of the students we flag, how many are really at risk. Recall: of all at-risk students, how many we flag
Baseline	A very simple method used for comparison, to show that a complex model is worth it
Overfitting	A model that memorised its training data and does badly on new data
Data leakage	Accidentally using information the model would not have in real life, which makes results look better than they are
Data drift	The data changes over time (new courses, new deadlines), so a model can get worse
Class imbalance	One outcome is much rarer than another
Clustering	Letting the computer group similar students without telling it the answer
Association rule	A pattern of the form "if A and B then C"; lift says how much more often C happens than normal
Effect size	How big a difference or link is, as opposed to whether it is statistically "significant"
p-value	The chance of seeing such a result by luck alone; with thousands of students, nearly everything gets a tiny p-value
Odds ratio	How many times higher the odds of an outcome are in one group than in another
Percentile rank	Where a value sits among others, from 0 (lowest) to 1 (highest)
Fairness (recall gap)	Whether the model finds at-risk students equally often in every group
Time series	Numbers recorded day by day, such as daily clicks

Word	Plain meaning
A/B test	A fair experiment: students are randomly given the help (A) or not (B), and the results are compared
Statistical power	The chance that an experiment will detect a real effect of a given size